

Streamlit Interface Testing

1. Artefact Overview

This artefact documents the Streamlit implementation of the phishing-email detection prototype. The supplied evidence contains 48 screenshot pages covering twelve email tests. Each test records four stages: email input, detection result, SHAP factors supporting the prediction, and SHAP factors working against the prediction together with the strongest SHAP factors.

The screenshots are functional and qualitative evidence. They demonstrate the implemented workflow but do not replace aggregate accuracy, precision, recall, F1-score or confusion-matrix evaluation.

2. How the Streamlit System Works

- 1) The user enters sender details, subject and email body and selects Analyze Email.
- 2) The application constructs the text representation from the subject and body and sends it for inference.
- 3) DeBERTa-v3-large generates a contextual text representation.
- 4) Twelve engineered phishing/structural features represent urgency, credential, threat, financial, call-to-action, URL, email-address, phone, exclamation, question, uppercase-ratio and text-length signals.
- 5) A learned gate combines the transformer representation with the engineered-feature representation before the two-class Safe/Phishing classifier produces probabilities.
- 6) Security Indicators and Email Characteristics are displayed separately as contextual evidence; they supplement the model and are not the final prediction itself.
- 7) When Explain AI Decision is selected, SHAP is run after inference to explain the already-produced model decision.

3. Understanding the Interface Output

3.1 Detection Result and Probability Breakdown

The Detection Result is the model's selected class. Confidence is the probability assigned to that selected class. Safe Probability and Phishing Probability show the two-class distribution for the individual email. These values describe model confidence for that email, not overall model accuracy.

3.2 Security Indicators and Email Characteristics

Security Indicators provide supplementary warnings such as suspicious call-to-action language or urgent/threatening language. A 'model-based detection' message means the learned contextual classifier identified phishing-like semantics even though no predefined indicator matched. Email Characteristics report neutral properties such as URL count, email-address presence and message length; these properties alone do not prove phishing.

4. SHAP Explainability Method

SHAP is post-hoc explainability: the model predicts first and SHAP then estimates which displayed tokens pushed the explained output in either direction. Generating a SHAP explanation does not change the class or probability.

Section	Meaning	Impact/interpretation
Factors Supporting the Prediction	Tokens contributing in the direction of the class being explained.	For a Phishing explanation they push toward Phishing; for a Safe explanation they push toward Safe.
Factors Working Against the Prediction	Tokens contributing in the opposite direction.	They provide counter evidence to the displayed explanation class.
Strongest SHAP Factors	Combined ranking of influential positive and negative contributions.	Shows the most influential local factors and their direction.
Contribution value	Direction and relative local influence.	Not a probability and not a universal risk score for that word.

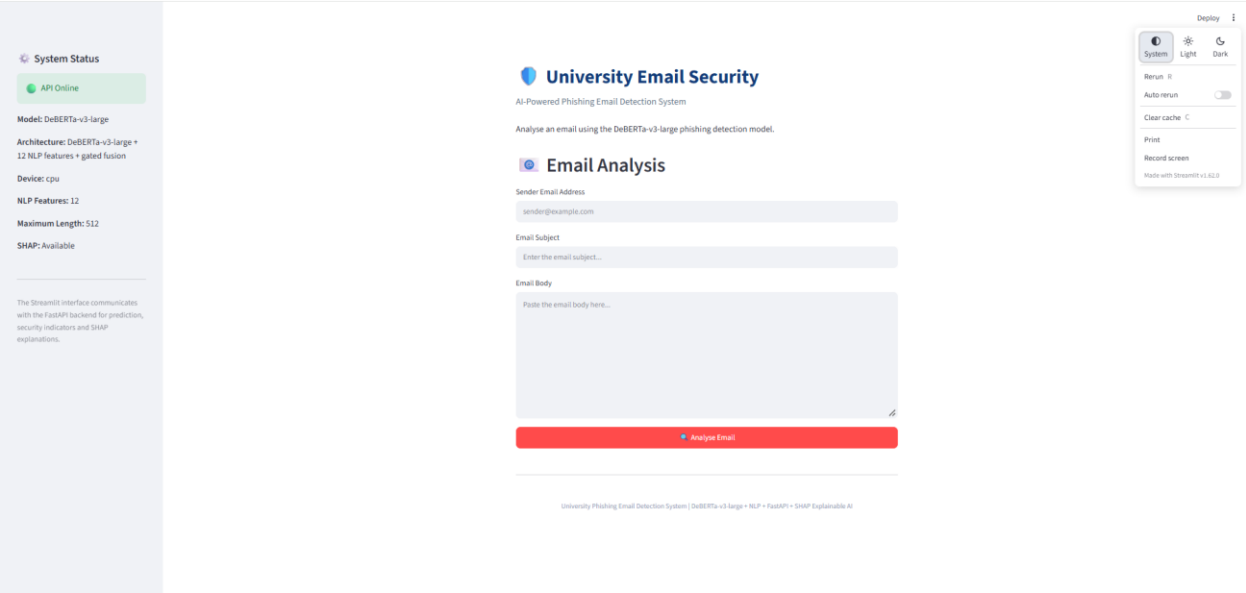
Ordinary words, punctuation, numbers and token fragments can appear because transformer models use contextual token representations. A word appearing as a supporting factor does not mean that word is inherently malicious; the attribution applies to that specific email and model output.

5. Evidence Summary

Test	Displayed result	Key evidence
13	Potential Phishing Email (99.97%)	Suspicious call-to-action; email address; phone; moderate length
14	Safe Email (99.98%)	No common phishing indicators; moderate length
15	Potential Phishing Email (99.97%)	Model-based detection; moderate length
16	Safe Email (99.99%)	No common phishing indicators; long-form
17	Potential Phishing Email (99.97%)	Model-based detection; moderate length
18	Safe Email (99.98%)	Urgent/threatening indicator; long-form
19	Potential Phishing Email (99.82%)	Model-based detection; 2 URLs/links; moderate length
20	Safe Email (99.98%)	No common phishing indicators; moderate length
21	Safe Email (54.33%)	Manual-review warning; urgency; URL; email address; long-form
22	Safe Email (99.99%)	No common phishing indicators; long-form
23	Potential Phishing Email (99.97%)	Model-based detection; long-form
24	Safe Email (99.98%)	No common phishing indicators; moderate length

Test 9 is particularly valuable because the model is close to the decision boundary: 54.33% Safe versus 45.67% Phishing. The interface therefore recommends manual review instead of presenting the result as highly certain.

6. Steamlit Interface



6.1 Streamlit Test 1

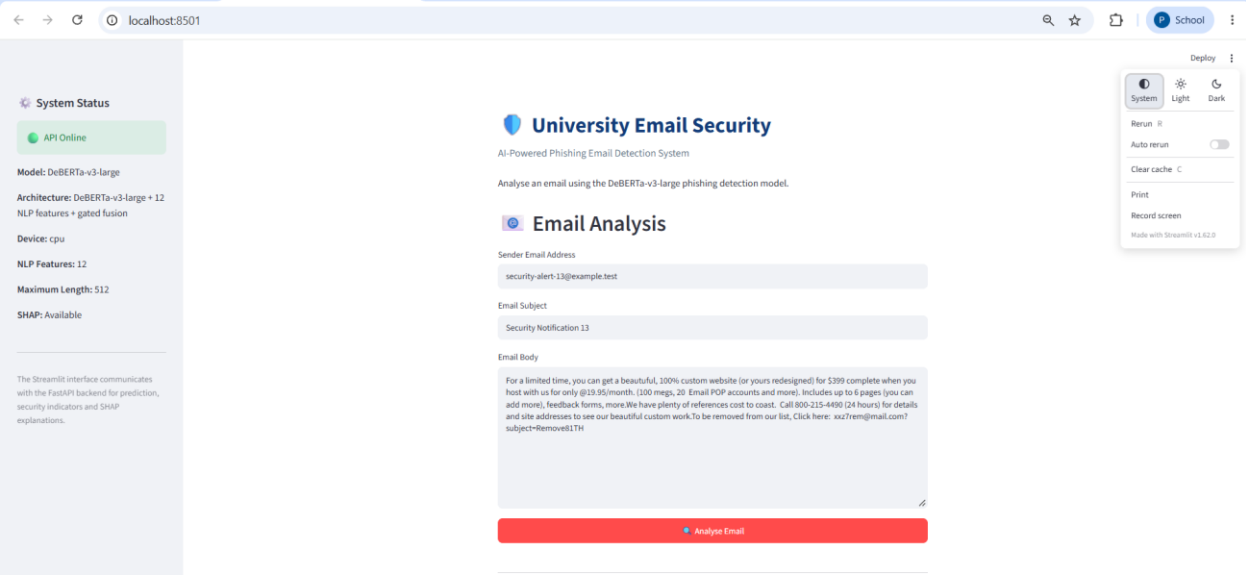


Figure S1. Test 1: Email input and submission.

The input page records Senders email address, subject and the submitted body.

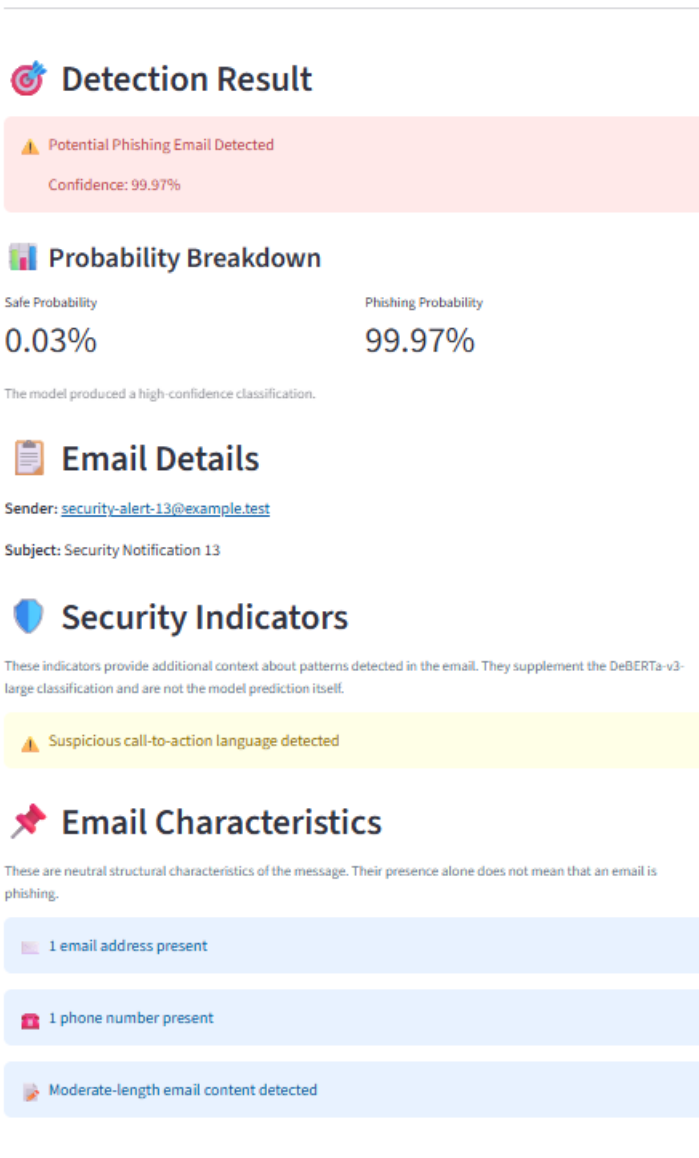


Figure S2. Test 1: Detection result and contextual indicators.

The model returns Potential Phishing Email at 99.97%. The call-to-action warning and structural characteristics are supplementary context.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Phishing Email

Factors Supporting the Prediction

1. to
Contribution: 0.085648
2. up
Contribution: 0.066845
3. 13
Contribution: 0.032188
4. time
Contribution: 0.028848
5. POP
Contribution: 0.006723
6. 100
Contribution: 0.006358
7. custom
Contribution: 0.006345
8. 95
Contribution: 0.005337
9. a
Contribution: 0.004983
10. can
Contribution: 0.004879

Figure S3. Test 1: SHAP factors supporting the prediction.

The SHAP explanation targets Phishing Email; positive factors therefore push the local output toward phishing.

Factors Working Against the Prediction

1. :
Contribution: -0.057442

2. {
Contribution: -0.051388

3. Includes
Contribution: -0.049838

4. .
Contribution: -0.031836

5. .
Contribution: -0.031382

6. month
Contribution: -0.017836

7. /
Contribution: -0.016689

8.)
Contribution: -0.013212

9. 100
Contribution: -0.008411

10. me
Contribution: -0.008323

Strongest SHAP Factors

1. to — 0.085640 (supports the phishing prediction)

2. up — 0.066045 (supports the phishing prediction)

3. : — -0.057442 (works against the phishing prediction)

4. { — -0.051388 (works against the phishing prediction)

5. Includes — -0.049838 (works against the phishing prediction)

6. 13 — 0.032100 (supports the phishing prediction)

7. . — -0.031836 (works against the phishing prediction)

8. . — -0.031382 (works against the phishing prediction)

9. time — 0.028840 (supports the phishing prediction)

10. month — -0.017836 (works against the phishing prediction)

Security Indicators Confirmed

 Suspicious call-to-action language detected

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S4. Test 1: Factors working against the prediction and strongest SHAP factors.

Negative factors push away from phishing. The strongest-factor list combines both directions and makes the evidence balance auditable.

6.2 Streamlit Test 2

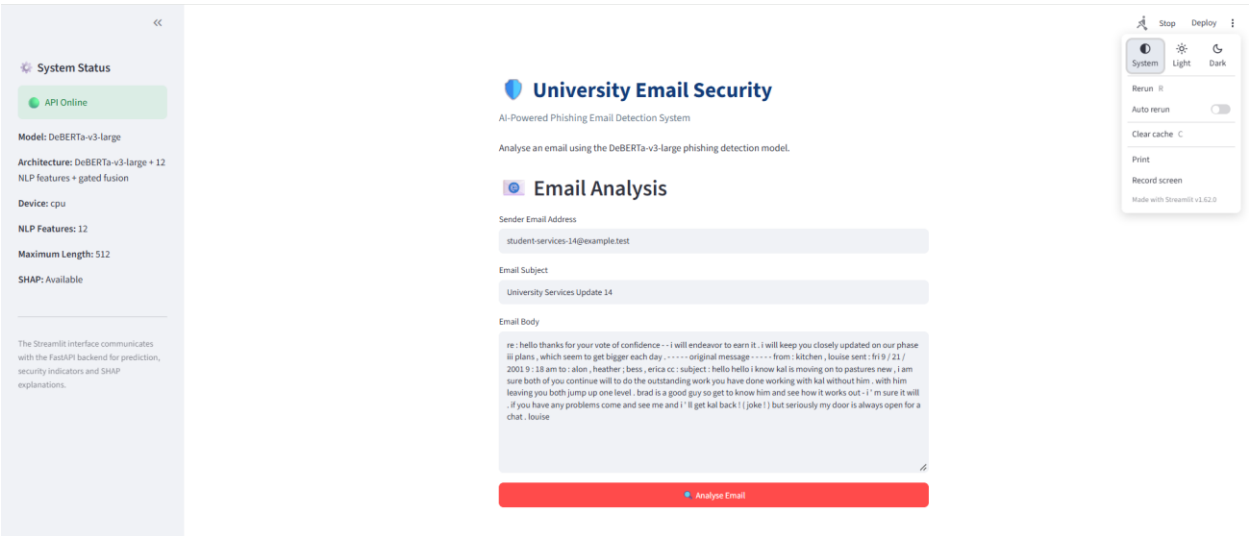


Figure S5. Test 2: Email input and submission.

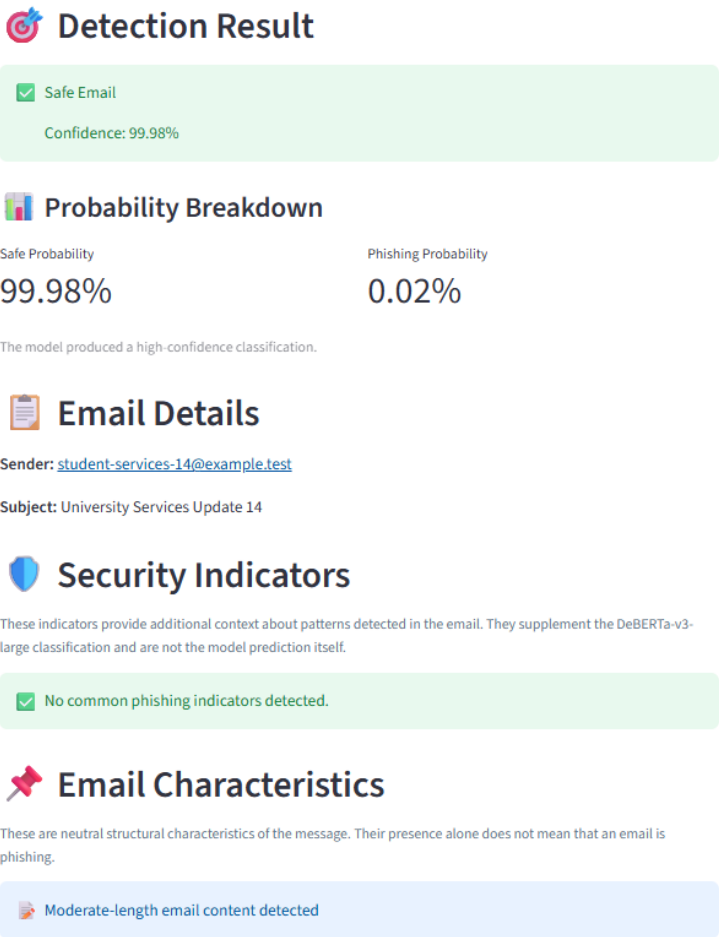


Figure S6. Test 2: Detection result and contextual indicators.

Safe Email is returned at 99.98%, with no common phishing indicators.

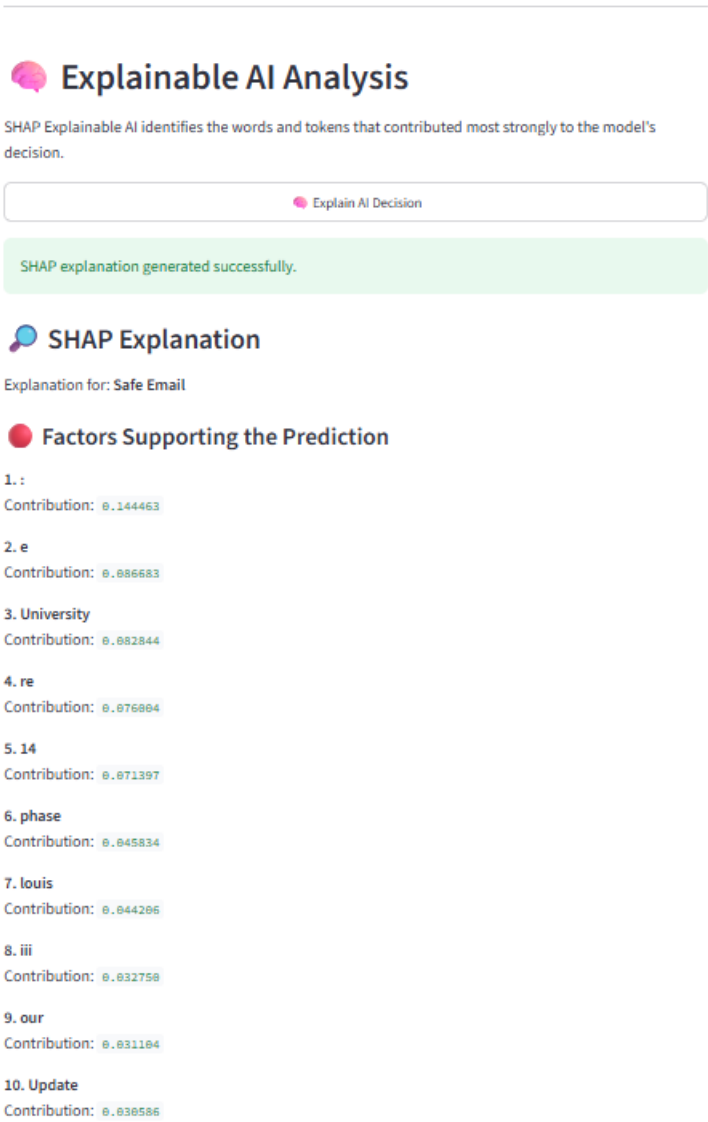


Figure S7. Test 2: SHAP factors supporting the prediction.

Because the explanation target is Safe Email, supporting factors push toward Safe. Visible examples include University and Update.

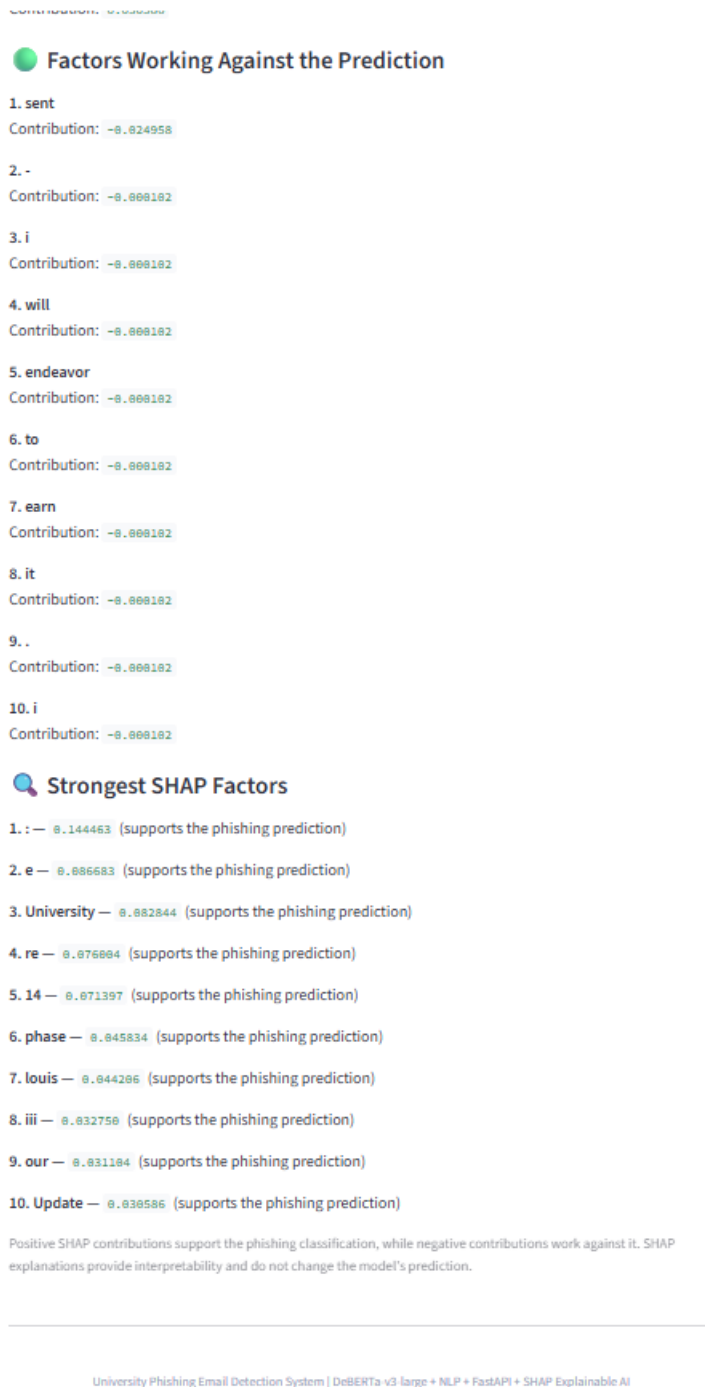


Figure S8. Test 2: Factors working against the prediction and strongest SHAP factors.

Working-against factors push away from Safe. Their presence shows that even a confident result can contain counter-evidence.

6.3 Streamlit Test 3

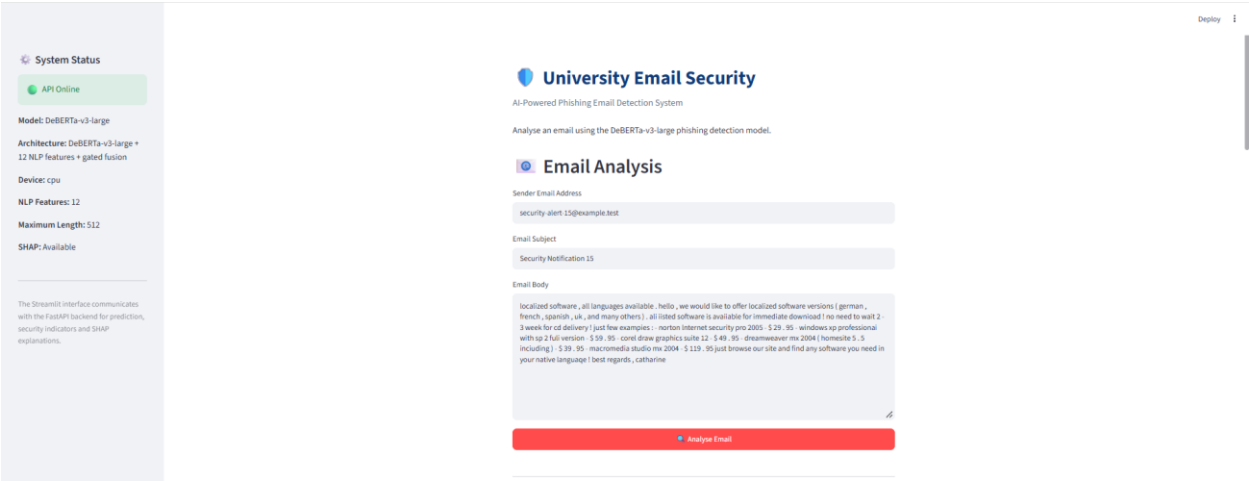


Figure S9. Test 3: Email input and submission.

The Streamlit form records security-alert-15@example.test, Security Notification 15 and the selected body.

Detection Result

 Potential Phishing Email Detected

Confidence: 99.97%

Probability Breakdown

Safe Probability

0.03%

Phishing Probability

99.97%

The model produced a high-confidence classification.

Email Details

Sender: security-alert-15@example.test

Subject: Security Notification 15

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

 Model-based detection

The model detected phishing based on contextual and semantic patterns. No predefined rule-based indicators matched this message.

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 Moderate-length email content detected

Figure S10. Test 3: Detection result and contextual indicators.

Potential Phishing Email is returned at 99.97%. 'Model-based detection' indicates contextual/semantic classification without a matching predefined rule.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Phishing Email

Factors Supporting the Prediction

1. Notification

Contribution: 0.006663

2. versions

Contribution: 0.006623

3. offer

Contribution: 0.005617

4. ,

Contribution: 0.004588

5. we

Contribution: 0.001949

6. a

Contribution: 0.001801

7. localized

Contribution: 0.001485

8. like

Contribution: 0.000919

9. software

Contribution: 0.000894

10. would

Contribution: 0.000889

Figure S11. Test 3: SHAP factors supporting the prediction.

Visible positive factors include Notification, versions and offer; they support phishing only in this local context.

● Factors Working Against the Prediction

1. Security
Contribution: -0.007795
2. to
Contribution: -0.004906
3. (
Contribution: -0.004046
4. many
Contribution: -0.003481
5. german
Contribution: -0.002422
6. .
Contribution: -0.002292
7. :
Contribution: -0.001333
8. and
Contribution: -0.000900
9.)
Contribution: -0.000876
10. french
Contribution: -0.000841

🔍 Strongest SHAP Factors

1. Security — -0.007795 (works against the phishing prediction)
2. Notification — 0.006663 (supports the phishing prediction)
3. versions — 0.006623 (supports the phishing prediction)
4. offer — 0.005617 (supports the phishing prediction)
5. to — -0.004906 (works against the phishing prediction)
6. , — 0.004588 (supports the phishing prediction)
7. (— -0.004046 (works against the phishing prediction)
8. many — -0.003481 (works against the phishing prediction)
9. german — -0.002422 (works against the phishing prediction)
10. . — -0.002292 (works against the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S12. Test 3: Factors working against the prediction and strongest SHAP factors.

Security and other tokens appear as counter-evidence. This demonstrates competing contextual signals rather than a keyword-only rule.

6.4 Streamlit Test 4

System Status

API Online

Model: DeBERTa-v3-large

Architecture: DeBERTa-v3-large + 12
NLP features + gated fusion

Device: cpu

NLP Features: 12

Maximum Length: 512

SHAP: Available

The Streamlit interface communicates with the FastAPI backend for prediction, security indicators and SHAP explanations.

Deploy

University Email Security

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

Email Analysis

Sender Email Address

student-services-16@example.test

Email Subject

University Services Update 16

Email Body

accounts for tolerate the only one we need is for jason sokolov , unless maureen ' s charges are for something that she does not know is tolerate . tanya tamarchenko does not need tolerate and she has the second largest percentage of the charges . anyway , the only tolerate subscription we need is for jason sokolov . reuters : (february charges : \$ 405 . 56) no one in research uses reuters . i believe most of the charges are for hector campos who used it when he was on the trading desk . when he rotated into the research group he did not need it any longer , but is still being billed for it . please remove from the research cost center . the following individuals are no longer with enron or no longer with research and their accounts should be removed from the research cost center . clayton vernon no longer with the research group remove his lim / lim / excel and lim core charges from the research cost center brad amoine no longer with enron remove his lim / lim / excel and lim core charges from the research cost center shalesh ganjoo no longer with the research group remove his lim and lim core charges from the research cost center i hope this is not too confusing i please advise . thanks ! shirley crenshaw

Analyse Email

Figure S13. Test 16: Email input and submission.

The input records student-services-16@example.test, University Services Update 16 and a long-form body.

Detection Result

✓ Safe Email

Confidence: 99.99%

Probability Breakdown

Safe Probability

99.99%

Phishing Probability

0.01%

The model produced a high-confidence classification.

Email Details

Sender: student-services-16@example.test

Subject: University Services Update 16

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

✓ No common phishing indicators detected.

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 Long-form email content detected

Figure S14. Test 4: Detection result and contextual indicators.

Safe Email is returned at 99.99%; no common phishing indicators are detected.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: **Safe Email**

Factors Supporting the Prediction

1. **that**
Contribution: 0.142671
2. **me**
Contribution: 0.111703
3. **told**
Contribution: 0.080500
4. **16**
Contribution: 0.048938
5. **reen**
Contribution: 0.042880
6. **:**
Contribution: 0.042293
7. **University**
Contribution: 0.041008
8. **tan**
Contribution: 0.037967
9. **Update**
Contribution: 0.035353
10. **ya**
Contribution: 0.034051

Figure S15. Test 4: SHAP factors supporting the prediction.

The Safe explanation shows supporting factors including that, me, told, University and Update.

● Factors Working Against the Prediction

1. when

Contribution: -0.000378

2. she

Contribution: -0.000378

🔍 Strongest SHAP Factors

1. that — 0.142671 (supports the phishing prediction)

2. me — 0.111703 (supports the phishing prediction)

3. told — 0.088500 (supports the phishing prediction)

4. 16 — 0.048938 (supports the phishing prediction)

5. reen — 0.042800 (supports the phishing prediction)

6. : — 0.042293 (supports the phishing prediction)

7. University — 0.041008 (supports the phishing prediction)

8. tan — 0.037967 (supports the phishing prediction)

9. Update — 0.035353 (supports the phishing prediction)

10. ya — 0.034051 (supports the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S16. Test 4: Factors working against the prediction and strongest SHAP factors.

The strongest-factor view is dominated by Safe-supporting evidence, consistent with the very high Safe probability.

6.5 Streamlit Test 5

Displayed outcome: Potential Phishing Email | Confidence: 99.97%. Model-based detection; moderate length.



University Email Security

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.



Email Analysis

Sender Email Address

security-alert-17@example.test

Email Subject

Security Notification 17

Email Body

american positive seam dramaturgy brumidi blather petition bay kemp hectic gradient out of energy . . . if so , i think i ' ve found the answer scorch esmark judas fondle feminism nameplate nonetheless quantitative peafowl fealty cinch exterminate cowan congo quest attache maya inhabitation council downstate expectant cockeye blythe ester isomorphic physique beplaster burg eave godmother flog andesine remittance physic degrease dugan amplifier convulsion danbury illusory griffith plywood osseous

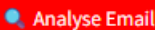


Figure S17. Test 5: Email input and submission.

The input records security-alert-17@example.test, Security Notification 17 and the submitted body.

Detection Result

 Potential Phishing Email Detected

Confidence: 99.97%

Probability Breakdown

Safe Probability

0.03%

Phishing Probability

99.97%

The model produced a high-confidence classification.

Email Details

Sender: security-alert-17@example.test

Subject: Security Notification 17

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

 Model-based detection

The model detected phishing based on contextual and semantic patterns. No predefined rule-based indicators matched this message.

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 Moderate-length email content detected

Figure S18. Test 5: Detection result and contextual indicators.

Potential Phishing Email is returned at 99.97% with model-based detection.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Phishing Email

Factors Supporting the Prediction

1. Notification

Contribution: 0.013324

2. es

Contribution: 0.003367

3. scorch

Contribution: 0.002308

4. positive

Contribution: 0.002081

5. 17

Contribution: 0.001877

6. as

Contribution: 0.001481

7. american

Contribution: 0.001391

8. seam

Contribution: 0.000698

9. nameplate

Contribution: 0.000342

10. le

Contribution: 0.000278

Figure S19. Test 5: SHAP factors supporting the prediction.

Visible supporting factors include Notification, research, positive and american; they locally push toward phishing.

Factors Working Against the Prediction

1. Security

Contribution: -0.015610

2. answer

Contribution: -0.005164

3. :

Contribution: -0.002609

4. fond

Contribution: -0.001044

5. found

Contribution: -0.000489

6. i

Contribution: -0.000483

7. think

Contribution: -0.000472

8. ve

Contribution: -0.000144

9. the

Contribution: -0.000024

Strongest SHAP Factors

1. Security — -0.015610 (works against the phishing prediction)

2. Notification — 0.013324 (supports the phishing prediction)

3. answer — -0.005164 (works against the phishing prediction)

4. es — 0.003367 (supports the phishing prediction)

5. : — -0.002609 (works against the phishing prediction)

6. scorch — 0.002300 (supports the phishing prediction)

7. positive — 0.002081 (supports the phishing prediction)

8. 17 — 0.001877 (supports the phishing prediction)

9. as — 0.001481 (supports the phishing prediction)

10. american — 0.001391 (supports the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S20. Test 5: Factors working against the prediction and strongest SHAP factors.

Security and other tokens work against phishing. The combined list shows both directions behind the high-confidence result.

6.6 Streamlit Test 6

Displayed outcome: Safe Email | Confidence: 99.98%. Urgent/threatening indicator; long-form.



University Email Security

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.



Email Analysis

Sender Email Address

student-services-18@example.test

Email Subject

University Services Update 18

Email Body

confirmed their participation as plenary lecturers . proposals of communications are welcome , with a deadline of may 31 , 1998 . papers will last 20 minutes with a 10 - minute discussion period to follow . abstracts should be no longer than 200 words and should be submitted in print and on a floppy disk (microsoft word or word perfect) to the following address : secretaria de selim98 facultade de humanidades . universidade de vigo aptdo . 874 . e-36200 vigo (spain) submissions should include name (s) and institutional affiliation (s) of the author (s) , telephone , fax and e-mail address . please indicate if you need an overhead projector , a computer or any other facilities . acceptance of proposals will be communicated in june 1998 . the third circular will be sent around june 1998 and will carry fuller information on the arrangements for the conference . for any enquiries about accomodation or travel arrangements please contact bives tour at info @ conbives . com . for further information visit the conference website at : http : / / www . uvigo . es / eventos / congresos / selim98 / or send and e-mail to : selim98 @ uvigo . es the organising committee , may 1998 ana m bringas lpez dolores gonzalez alvarez javier prez guerra eduardo varela bravo

Analyse Email

Figure S21. Test 6: Email input and submission.

The input records student-services-18@example.test, University Services Update 18 and a long academic/conference-style body.

Detection Result

 Safe Email

Confidence: 99.98%

Probability Breakdown

Safe Probability

99.98%

Phishing Probability

0.02%

The model produced a high-confidence classification.

Email Details

Sender: student-services-18@example.test

Subject: University Services Update 18

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

 Urgent or threatening language detected

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 Long-form email content detected

Figure S22. Test 6: Detection result and contextual indicators.

Safe Email is returned at 99.98% even though the separate rule layer flags urgent/threatening language.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Safe Email

Factors Supporting the Prediction

1. University

Contribution: 0.167469

2. :

Contribution: 0.147329

3. 18

Contribution: 0.118988

4. medieval

Contribution: 0.067368

5. Update

Contribution: 0.059822

6. :

Contribution: 0.056547

7. -

Contribution: 0.041998

8. xi

Contribution: 0.039836

9. .

Contribution: 0.031424

10. :

Contribution: 0.030469

Figure S23. Test 6: SHAP factors supporting the prediction.

The Safe explanation includes local supporting tokens such as University, medieval and Update.

Factors Working Against the Prediction

1. **xi**
Contribution: -0.031264
2. **in**
Contribution: -0.029338
3. **Services**
Contribution: -0.010064
4. **Subject**
Contribution: -0.000227

Strongest SHAP Factors

1. **University** — 0.167409 (supports the phishing prediction)
2. **:** — 0.147329 (supports the phishing prediction)
3. **18** — 0.110908 (supports the phishing prediction)
4. **medieval** — 0.067368 (supports the phishing prediction)
5. **Update** — 0.059822 (supports the phishing prediction)
6. **:** — 0.056547 (supports the phishing prediction)
7. **-** — 0.041990 (supports the phishing prediction)
8. **xi** — 0.039836 (supports the phishing prediction)
9. **.** — 0.031424 (supports the phishing prediction)
10. **xi** — -0.031264 (works against the phishing prediction)

Security Indicators Confirmed

 Urgent or threatening language detected

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S24. Test 6: Factors working against the prediction and strongest SHAP factors.

This case is important: the urgency warning does not override the learned classifier. SHAP shows the token-level evidence behind the Safe result.

6.7 Detailed Streamlit Test 7

Displayed outcome: Potential Phishing Email | Confidence: 99.82%. Model-based detection; 2 URLs/links; moderate length.

 **University Email Security**

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

 **Email Analysis**

Sender Email Address

security-alert-19@example.test

Email Subject

Security Notification 19

Email Body

Hi Job Seeker,When you create a FREE My Net-Temps account, you have access to the tools and resources for finding your next job more effectively. It only takes a minute. Plus you get news and tips specific to your profession.Everything you need is in your My Net-Temps account!* Post up to 3 resumes with the Build My Resume and Copy & Paste tools

- * Setup custom search agents to automatically receive job leads
- * Resume statistics to track your success
- * Resume, cover letter and thank you letter writing tips
- * Salary calculator
- * Career articles and weekly newsletter
- * Access to over 7500 recruiters that can help your job searchSetup your account now at <http://www.net-temps.com/careerdev/>The Net-Temps Team

www.net-temps.com

 Analyse Email

Figure S25. Test 7: Email input and submission.

The input records security-alert-19@example.test, Security Notification 19 and a job/resource-themed body containing links.

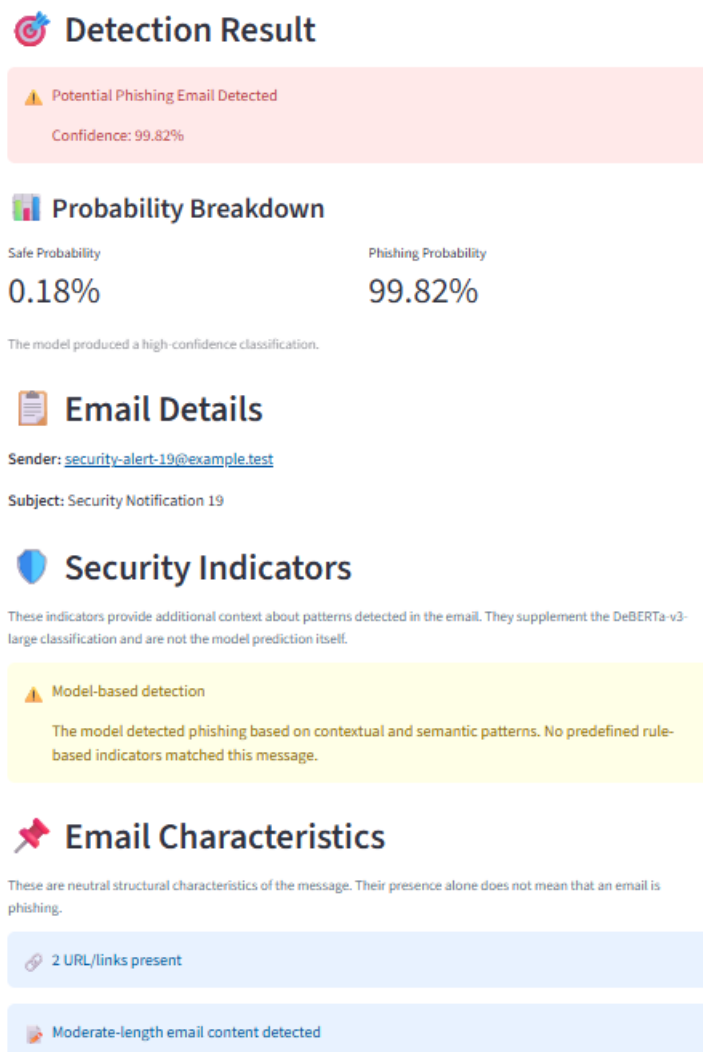


Figure S26. Test 7: Detection result and contextual indicators.

Potential Phishing Email is returned at 99.82%; two URLs/links are also reported.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Phishing Email

Factors Supporting the Prediction

1. Notification
Contribution: 0.006663
2. When
Contribution: 0.008825
3. 19
Contribution: 0.006687
4. ,
Contribution: 0.00552
5. Hi
Contribution: 0.00464
6. Job
Contribution: 0.00359
7. Seeker
Contribution: 0.00336
8. Plus
Contribution: 0.00208
9. need
Contribution: 0.00173
10. next
Contribution: 0.00158

Figure S27. Test 7: SHAP factors supporting the prediction.

Visible supporting factors include Notification, When, Job, Seeker and Plus.

Factors Working Against the Prediction

1. Security

Contribution: -0.007884

2. :

Contribution: -0.001338

3. you

Contribution: -0.000029

4. get

Contribution: -0.000029

5. news

Contribution: -0.000028

6. and

Contribution: -0.000028

7. resources

Contribution: -0.000023

8. Temp

Contribution: -0.000001

Strongest SHAP Factors

1. Security — -0.007884 (works against the phishing prediction)

2. Notification — 0.006663 (supports the phishing prediction)

3. : — -0.001338 (works against the phishing prediction)

4. When — 0.000825 (supports the phishing prediction)

5. 19 — 0.000687 (supports the phishing prediction)

6. , — 0.000552 (supports the phishing prediction)

7. Hi — 0.000464 (supports the phishing prediction)

8. Job — 0.000359 (supports the phishing prediction)

9. Seeker — 0.000336 (supports the phishing prediction)

10. Plus — 0.000208 (supports the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S28. Test 7: Factors working against the prediction and strongest SHAP factors.

Security and other message tokens oppose phishing. Strongest SHAP Factors ranks both positive and negative local evidence.

6.8 Streamlit Test 8

Displayed outcome: Safe Email | Confidence: 99.98%. No common phishing indicators; moderate length.

Figure S29. Test 8: Email input and submission.

The input records student-services-20@example.test, University Services Update 20 and an application/resource-request message.

 **University Email Security**

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

 **Email Analysis**

Sender Email Address

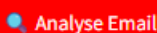
student-services-20@example.test

Email Subject

University Services Update 20

Email Body

please approve : application request (wsmh - 4 esnva) security resource request wsmh - 4 esnva has been submitted for your approval . to view the request , double click your left mouse button on the notes document link below . quick steps to approve or reject request form : 1 . click the button to view details of the requests . 2 . click the button to approve or reject the requests . 3 . to edit the request form , double - click anywhere on the form . see the online help for instructions or call ect security .

 Analyse Email

Detection Result

 Safe Email

Confidence: 99.98%

Probability Breakdown

Safe Probability

99.98%

Phishing Probability

0.02%

The model produced a high-confidence classification.

Email Details

Sender: student-services-20@example.test

Subject: University Services Update 20

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

 No common phishing indicators detected.

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 Moderate-length email content detected

Figure S30. Test 8: Detection result and contextual indicators.

Safe Email is returned at 99.98% with no common phishing indicators.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Safe Email

Factors Supporting the Prediction

1. document
Contribution: 0.077646
2. the
Contribution: 0.071308
3. notes
Contribution: 0.071308
4. quick
Contribution: 0.069387
5. link
Contribution: 0.065959
6. Services
Contribution: 0.055769
7. :
Contribution: 0.049453
8. (
Contribution: 0.046731
9. University
Contribution: 0.040589
10. below
Contribution: 0.040069

Figure S31. Test 8: SHAP factors supporting the prediction.

The Safe explanation includes document, notes, quick, link, Services and University among supporting factors.

Factors Working Against the Prediction

1. approval
Contribution: -0.024332
2. to
Contribution: -0.015463
3. request
Contribution: -0.009875
4. n
Contribution: -0.004843
5. va
Contribution: -0.004843
6. has
Contribution: -0.004843
7. :
Contribution: -0.000799
8. application
Contribution: -0.000799
9. steps
Contribution: -0.000310
10. to
Contribution: -0.000310

Strongest SHAP Factors

1. document — 0.077646 (supports the phishing prediction)
2. the — 0.071308 (supports the phishing prediction)
3. notes — 0.071308 (supports the phishing prediction)
4. quick — 0.069387 (supports the phishing prediction)
5. link — 0.065959 (supports the phishing prediction)
6. Services — 0.055769 (supports the phishing prediction)
7. : — 0.049453 (supports the phishing prediction)
8. (— 0.046731 (supports the phishing prediction)
9. University — 0.040589 (supports the phishing prediction)
10. below — 0.040069 (supports the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

University Phishing Email Detection System | DeBERTa v3-large + NLP + FastAPI + SHAP Explainable AI

Figure S32. Test 8: Factors working against the prediction and strongest SHAP factors.

Approval, request and application appear among opposing factors, showing that suspicious-looking vocabulary can exist without overturning the overall Safe result.

6.9 Streamlit Test 9

Displayed outcome: Safe Email | Confidence: 54.33%. Manual-review warning; urgency; URL; email address; long-form.



University Email Security

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

Email Analysis

Sender Email Address

security-alert-21@example.test

Email Subject

Security Notification 21

Email Body

irons in the fire too. That only happens by being in all the right places at all the right times. So even if you do it by hand, get maximum exposure and be found on as many career sites as possible.Just food for thought!Best Wishes,Elizabeth Bradley Manager
elizabeth-bradley@allexecs.org=====The staff of AllExecs.org have years of recruiting experience and may from time to time send you a review of a career tool that has helped others in their job search. If you'd rather not receive these reviews, reply to this note with "Remove" in the subject line.=====

 Analyse Email

Figure S33. Test 9: Email input and submission.

The input records security-alert-21@example.test, Security Notification 21 and a recruitment/job-review style body.

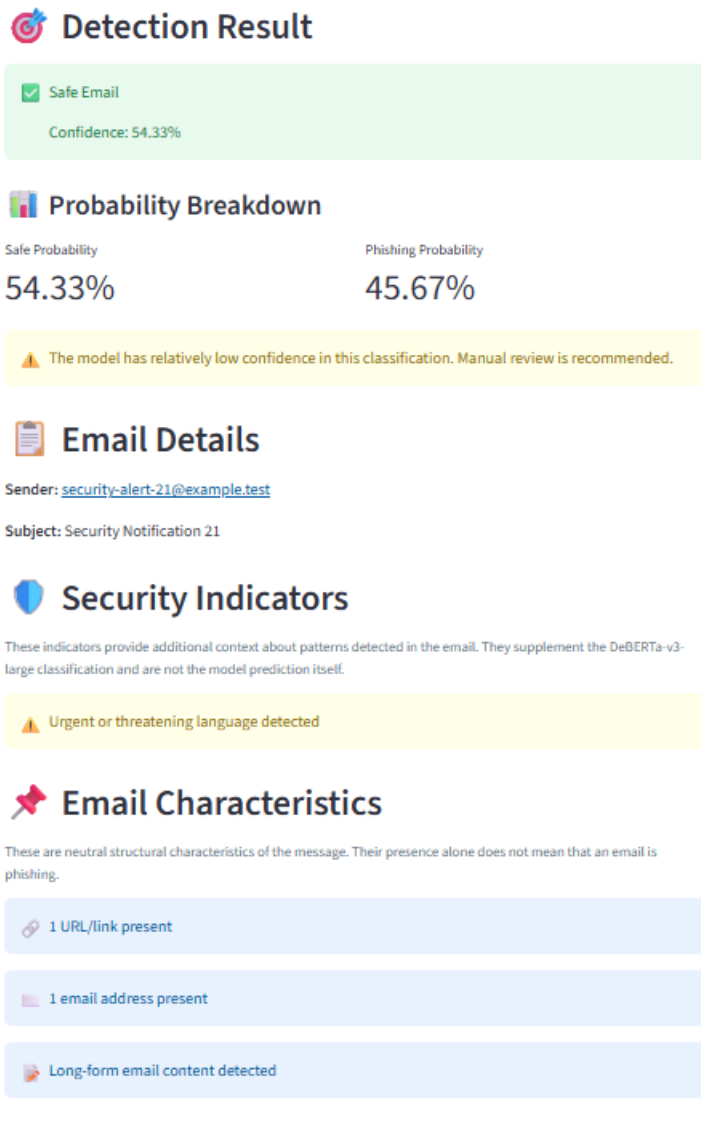


Figure S34. Test 9: Detection result and contextual indicators.

Safe Email is returned at only 54.33% versus 45.67% Phishing. The interface explicitly recommends manual review and reports urgency, URL, email-address and long-form characteristics.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: **Safe Email**

Factors Supporting the Prediction

1. **a**
Contribution: 0.858821
2. **across**
Contribution: 0.849514
3. **copy**
Contribution: 0.839846
4. **of**
Contribution: 0.838973
5. **your**
Contribution: 0.832785
6. **the**
Contribution: 0.831553
7. **on**
Contribution: 0.825557
8. **Security**
Contribution: 0.822795
9. **I**
Contribution: 0.814978
10. **internet**
Contribution: 0.818222

Figure S35. Test 9: SHAP factors supporting the prediction.

The SHAP explanation targets Safe, so the supporting factors push toward the displayed Safe result; their interpretation must be considered alongside the near-balanced probabilities.

● Factors Working Against the Prediction

1. .
Contribution: -0.067487
2. Notification
Contribution: -0.067448
3. I
Contribution: -0.037474
4. :
Contribution: -0.035254
5. resume
Contribution: -0.027142
6. There
Contribution: -0.021953
7. 21
Contribution: -0.020203
8. recently
Contribution: -0.019844
9. Hi
Contribution: -0.018829
10. ,
Contribution: -0.016731

🔍 Strongest SHAP Factors

1. . — -0.067487 (works against the phishing prediction)
2. Notification — -0.067448 (works against the phishing prediction)
3. a — 0.050821 (supports the phishing prediction)
4. across — 0.049514 (supports the phishing prediction)
5. copy — 0.039846 (supports the phishing prediction)
6. of — 0.038973 (supports the phishing prediction)
7. I — -0.037474 (works against the phishing prediction)
8. : — -0.035254 (works against the phishing prediction)
9. your — 0.032705 (supports the phishing prediction)
10. the — 0.031553 (supports the phishing prediction)

🔒 Security Indicators Confirmed

⚠️ Urgent or threatening language detected

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S36. Test 9: Factors working against the prediction and strongest SHAP factors.

Several factors oppose Safe, consistent with uncertainty. The Streamlit screenshot does not display ground truth, so this case should be cross-checked against the held-out label before calling it correct or incorrect.

6.10 Streamlit Test 10

Displayed outcome: Safe Email | Confidence: 99.99%. No common phishing indicators; long-form.

 **University Email Security**

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

 **Email Analysis**

Sender Email Address

student-services-22@example.test

Email Subject

University Services Update 22

Email Body

> > something wrong.
> >
> > Maybe we should really go dirty Tleilaxu all the way.
> >
> >
> > Maybe y'all should buy m-w some more bandwidth.
>
> m-w?
>
merriam-webster. (:
C--
"I don't take no stocks in mathematics, anyway" --Huckleberry Finn

Analyse Email

☐ Analysing email with DeBERTa-v3-large...

Figure S37. Test 10: Email input and submission.

The input records student-services-22@example.test, University Services Update 22 and a long quoted/discussion-style body.

Detection Result

 Safe Email

Confidence: 99.99%

Probability Breakdown

Safe Probability

99.99%

Phishing Probability

0.01%

The model produced a high-confidence classification.



Email Details

Sender: student-services-22@example.test

Subject: University Services Update 22



Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.



No common phishing indicators detected.



Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.



Long-form email content detected

Figure S38. Test 10: Detection result and contextual indicators.

Safe Email is returned at 99.99%, with no common phishing indicators.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Safe Email

Factors Supporting the Prediction

1. DNA

Contribution: 0.122275

2. >

Contribution: 0.185858

3. .

Contribution: 0.075998

4. s

Contribution: 0.074478

5. idea

Contribution: 0.069468

6. >

Contribution: 0.069435

7. .

Contribution: 0.042886

8. >

Contribution: 0.042886

9. >

Contribution: 0.036862

10. University

Contribution: 0.035558

Figure S39. Test 10: SHAP factors supporting the prediction.

The Safe explanation shows tokens such as DNA, idea and University among local supporting factors.

Factors Working Against the Prediction

1.:

Contribution: -0.007101

2.:

Contribution: -0.000094

Strongest SHAP Factors

1. DNA — 0.122275 (supports the phishing prediction)

2. > — 0.105858 (supports the phishing prediction)

3. : — 0.075990 (supports the phishing prediction)

4. s — 0.074478 (supports the phishing prediction)

5. idea — 0.069460 (supports the phishing prediction)

6. > — 0.069435 (supports the phishing prediction)

7. : — 0.042006 (supports the phishing prediction)

8. > — 0.042006 (supports the phishing prediction)

9. > — 0.036062 (supports the phishing prediction)

10. University — 0.035558 (supports the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S40. Test 10: Factors working against the prediction and strongest SHAP factors.

Only limited opposing evidence is visible; the strongest list is dominated by factors supporting the displayed Safe result.

6.11 Streamlit Test 11

Displayed outcome: Potential Phishing Email | Confidence: 99.97%. Model-based detection; long-form.

 **University Email Security**

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

 **Email Analysis**

Sender Email Address

security-alert-23@example.test

Email Subject

Security Notification 23

Email Body

contact of the security firm so that you will contact them and arrange how to take possession of the two trunk boxes as my late father ' s foreign partner . you shall open account in your country where the money will be lodged into while you make arrange to sell of sealed box containing the 652 caract of blue rough cut diamond . the security company do not know the content of what is in both boxes and do not disclose this to them . after you have transferred ownership of the two trunk boxes , from the security company to your country , you shall then make arrangement for me and my kid sister to come over to your country and start a new life . upon request by you i can send to you the certificate of deposit for your assurance . we are willing to conceed up to 25 % of the total sum to you for all the anticipated help from you hoping that it will cover all your running expenses before and after the transaction . however , we will desire that you will help us invest our share till we will be able to meet you up in your country for further talks , please if this proposal is ok by you , please write me back via this e - mail address (stansankoh @ netscape . net) so that i can give you all the necessary informations . best regards stanley sankoh email : stansankoh @ netscape . net

 **Analyse Email**

Figure S41. Test 11: Email input and submission.

The input records security-alert-23@example.test, Security Notification 23 and a long narrative body.

Detection Result

 Potential Phishing Email Detected

Confidence: 99.97%

Probability Breakdown

Safe Probability

0.03%

Phishing Probability

99.97%

The model produced a high-confidence classification.

Email Details

Sender: security-alert-23@example.test

Subject: Security Notification 23

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

 Model-based detection

The model detected phishing based on contextual and semantic patterns. No predefined rule-based indicators matched this message.

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 Long-form email content detected

Figure S42. Test 11: Detection result and contextual indicators.

Potential Phishing Email is returned at 99.97% with model-based detection.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Phishing Email

Factors Supporting the Prediction

1. **by**
Contribution: 0.067578
2. **president**
Contribution: 0.067578
3. **a**
Contribution: 0.040617
4. **23**
Contribution: 0.033001
5. **attention**
Contribution: 0.033001
6. **eldest**
Contribution: 0.026822
7. **son**
Contribution: 0.026822
8. **reply**
Contribution: 0.024309
9. **to**
Contribution: 0.023059
10. **jan**
Contribution: 0.022767

Figure S43. Test 11: SHAP factors supporting the prediction.

The phishing explanation includes by, president, attention, eldest, son and reply among supporting factors.

Factors Working Against the Prediction

1. `)`
Contribution: `-0.108740`
2. `net`
Contribution: `-0.100742`
3. `i`
Contribution: `-0.097298`
4. `san`
Contribution: `-0.088501`
5. `koh`
Contribution: `-0.088501`
6. `scape`
Contribution: `-0.056165`
7. `.`
Contribution: `-0.056165`
8. `stan`
Contribution: `-0.052968`
9. `net`
Contribution: `-0.023917`
10. `@`
Contribution: `-0.022813`

Strongest SHAP Factors

1. `)` — `-0.108740` (works against the phishing prediction)
2. `net` — `-0.100742` (works against the phishing prediction)
3. `i` — `-0.097298` (works against the phishing prediction)
4. `san` — `-0.088501` (works against the phishing prediction)
5. `koh` — `-0.088501` (works against the phishing prediction)
6. `by` — `0.067578` (supports the phishing prediction)
7. `president` — `0.067578` (supports the phishing prediction)
8. `scape` — `-0.056165` (works against the phishing prediction)
9. `.` — `-0.056165` (works against the phishing prediction)
10. `stan` — `-0.052968` (works against the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S44. Test 11: Factors working against the prediction and strongest SHAP factors.

Several fragments/words oppose phishing. Their presence illustrates that a high-confidence prediction can still contain counter-evidence.

6.12 Streamlit Test 12

Displayed outcome: Safe Email | Confidence: 99.98%. No common phishing indicators; moderate length.

 **University Email Security**

AI-Powered Phishing Email Detection System

Analyse an email using the DeBERTa-v3-large phishing detection model.

 **Email Analysis**

Sender Email Address

student-services-24@example.test

Email Subject

University Services Update 24

Email Body

immigration update louise and marcus , we ' re still in the waiting period with the green card process . as a reminder , we ' ll need the baby ' s birth certificate from houston vital statistics to start the process of obtaining a us passport and dual citizenship . bobbi has some helpful information on traveling with a baby which she will bring over for you . the compensation team is looking at the us plan issues related to pwcs deferred compensation scheme . try to send over a copy of louise ' s new social security card this week . also , we would like to follow up on your unreimbursed medical claims . regards ,

 Analyse Email

Figure S45. Test 12: Email input and submission.

The input records student-services-24@example.test, University Services Update 24 and an immigration-update style body.

Detection Result

 Safe Email

Confidence: 99.98%

Probability Breakdown

Safe Probability

99.98%

Phishing Probability

0.02%

The model produced a high-confidence classification.

Email Details

Sender: student-services-24@example.test

Subject: University Services Update 24

Security Indicators

These indicators provide additional context about patterns detected in the email. They supplement the DeBERTa-v3-large classification and are not the model prediction itself.

 No common phishing indicators detected.

Email Characteristics

These are neutral structural characteristics of the message. Their presence alone does not mean that an email is phishing.

 Moderate-length email content detected

Figure S46. Test 12: Detection result and contextual indicators.

Safe Email is returned at 99.98%, with no common phishing indicators.

Explainable AI Analysis

SHAP Explainable AI identifies the words and tokens that contributed most strongly to the model's decision.

 Explain AI Decision

SHAP explanation generated successfully.

SHAP Explanation

Explanation for: Safe Email

Factors Supporting the Prediction

1. University

Contribution: 0.067828

2. :

Contribution: 0.066549

3. e

Contribution: 0.062111

4. and

Contribution: 0.061348

5. houston

Contribution: 0.056556

6. vital

Contribution: 0.056556

7. us

Contribution: 0.051714

8. ,

Contribution: 0.051495

9. marc

Contribution: 0.049521

10. need

Contribution: 0.048979

Figure S47. Test 12: SHAP factors supporting the prediction.

The Safe explanation includes University, houston, vital, us, marc and need among supporting factors.

Factors Working Against the Prediction

1. **we**
Contribution: -0.017314
2. **immigration**
Contribution: -0.000799
3. **update**
Contribution: -0.000799
4. **louis**
Contribution: -0.000799
5. **bi**
Contribution: -0.000178
6. **has**
Contribution: -0.000178
7. **some**
Contribution: -0.000178
8. **statistics**
Contribution: -0.000006
9. **to**
Contribution: -0.000006
10. **start**
Contribution: -0.000006

Strongest SHAP Factors

1. **University** — 0.067020 (supports the phishing prediction)
2. **:** — 0.066549 (supports the phishing prediction)
3. **e** — 0.062111 (supports the phishing prediction)
4. **and** — 0.061348 (supports the phishing prediction)
5. **houston** — 0.056556 (supports the phishing prediction)
6. **vital** — 0.056556 (supports the phishing prediction)
7. **us** — 0.051714 (supports the phishing prediction)
8. **,** — 0.051495 (supports the phishing prediction)
9. **marc** — 0.049521 (supports the phishing prediction)
10. **need** — 0.048979 (supports the phishing prediction)

Positive SHAP contributions support the phishing classification, while negative contributions work against it. SHAP explanations provide interpretability and do not change the model's prediction.

Figure S48. Test 12: Factors working against the prediction and strongest SHAP factors.

Immigration, update, louis, statistics and start appear among factors working against Safe. The strongest list summarises the local evidence without changing the prediction.

7. Cross-Test Findings

- Both Safe and Potential Phishing outcomes are demonstrated through the same Streamlit workflow.
- High-confidence examples show clear model preference, while Test 9 demonstrates an uncertain/borderline result and manual-review behaviour.
- Rule-derived indicators are not treated as the classifier itself. Test 18 is strong evidence: urgency is flagged but the model still strongly favours Safe.
- Model-based detection demonstrates that contextual semantic patterns can drive a phishing verdict even when predefined indicators do not match.
- SHAP consistently exposes both supporting and opposing local evidence, improving transparency over a probability-only interface.
- Strongest SHAP Factors provides a concise combined ranking useful for technical review and demonstration.

8. Limitations and Evidence Boundaries

- The screenshots are qualitative examples and must not be converted into an overall accuracy percentage.
- The Streamlit screenshots do not show dataset ground truth; labels should be cross-checked with the held-out dataset when correctness is being claimed.
- SHAP contributions are local explanations, not causal proofs, probabilities or universal phishing-word scores.
- Security indicators are supplementary contextual rules/features and are not the final model prediction.
- Sender addresses in this controlled test are synthetic metadata; the classifier's text input is based on the constructed subject/body representation.
- Long emails are subject to the classifier token limit, and SHAP uses a smaller explanation configuration for computational tractability.
- Borderline outputs such as Test 9 require human review and demonstrate residual model uncertainty.

9. Conclusion

This artefact demonstrates an operational Streamlit phishing-email analysis workflow integrating user input, DeBERTa-v3-large gated-hybrid inference, probability reporting, supplementary security indicators and SHAP explainability. Across above tests, the evidence shows both prediction classes, semantic model-based detections, rule/model disagreement and an explicit low-confidence manual-review case.